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Scientific Initiation Project

Prediction of Specialist Functional Assessment in Return to Sport Post-ACL Reconstruction Using Markerless and Machine Learning Approaches Through a Unipodal Multiplanar Plyometric Test

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Abstract

Prediction of Specialist Functional Assessment in ACL Injury Using Markerless and Machine Learning Approaches Through a Multiplanar Unipodal Plyometric Test

This project aims to apply Machine Learning models and Computer Vision techniques to estimate functional assessments made by specialists for patients with Anterior Cruciate Ligament (ACL) injuries, using kinematic data obtained from markerless tracking during the Multiplanar Unipodal Plyometric (CUBE) test. Three GoPro Hero 10 cameras will capture movements in multiple planes, and the MediaPipe framework will extract relevant kinematic features associated with ACL injury risk factors. Specialists will assign global functional scores (1–5) to patients, based on video analysis and their prior clinical knowledge. Inter-rater reliability will be evaluated using Cohen's Kappa index. The resulting data will be used to train and evaluate multiple supervised machine learning models (Extreme Gradient Boosting, Random Forest, Logistic Regression, K-Nearest Neighbors, Support Vector Machine, Gaussian Naive Bayes, Multilayer Perceptron), optimized with Grid Search and evaluated by stratified cross-validation. Model performance will be assessed by balanced accuracy, precision, recall, and F1 Score. Statistical analyses, performed in Python 3.12.11, will include normality (Shapiro-Wilk), homogeneity (Levene), group comparisons by one-way ANOVA or Kruskal-Wallis, appropriate post-hoc tests, and effect size estimation (Cohen's d). The ultimate goal is to develop an automated, objective, and reliable tool to support functional rehabilitation and clinical decision-making for post-ACL reconstruction patients and inidicates the best machine learning models for this task.

***Keyword**: Knee biomechanics, Functional rehabilitation, ACL reconstruction, Computer vision, Machine Learning*

1 Introduction

The Anterior Cruciate Ligament (ACL) plays a crucial role in knee stabilization, assisting in maintaining dynamic-static stability, joint coordination, limiting anterior displacement of the tibia relative to the femur, and also participating in resistance to internal rotation [1]. Its injury presents a high global incidence, corresponding to a significant portion of injuries affecting the knee joint, reported in some studies [1, 2] as responsible for approximately half of these cases, which makes it one of the main challenges in orthopedic and sports practice.

Among the factors contributing to ACL injury are biomechanical mechanisms related to inadequate movement patterns during sports and functional activities. Although the reason for the injury is multifactorial, there is consensus in the literature on specific movement patterns that significantly increase ligament vulnerability, such as the combination of knee valgus with low ankle, trunk, and knee flexion [3, 4, 5, 6, 7, 8, 9]. Positions close to joint extension also decrease the capacity for dissipation of vertical and ground reaction forces, which favors an increase in load on the knee—particularly on the ACL—elevating the risk of injury [10].

The rehabilitation process post-ACL injury and reconstruction aims to restore knee function, overcome psychological barriers, prevent additional injuries and recurrence, and optimize long-term quality of life. Although different therapeutic approaches are proposed, there is consensus on the importance of evidence-based practice. This approach involves constant evaluation of the recovery status and structured exercise prescription, including resistance training, neuromuscular exercises, execution of high-level functional tasks, and specific activities aimed at individual functionality, such as directed sports training [11]. Multiplanar plyometric tests, such as single-leg or depth jumps, have gained increasing relevance in the context of neuromuscular reconditioning and functional rehabilitation as they are classified as high-level dynamic functional tasks. Current literature highlights their importance in improving movement patterns, identifying risk factors, and allowing the development of efficient motor strategies [12].

Plyometric exercises consist of high-power movements that use the stretch-shortening cycle (SSC), a phenomenon where the eccentric phase—characterized by muscle activation during stretching—allows for the accumulation of elastic energy in the passive and contractile elements of the musculotendinous system. During this phase, structures like the titin protein contribute significantly to the generated force with low energy demand, due to the stiffness increase induced by muscle activation and controlled stretching [13]. Subsequently, in the concentric phase, this stored energy is released to amplify force production more efficiently. The capacity of muscle and connective tissues to store and reuse energy, as described by Herbert and Gandevia (2019), depends on their passive mechanical properties, such as viscoelasticity and muscle resting length [14]. This synergy between the eccentric and concentric phases is widely recognized as a central mechanism for both improving athletic performance and effective musculoskeletal rehabilitation strategies, especially when intensity progression is included. This progression prepares the musculoskeletal system for rapid movements and high forces that may be similar to the demands imposed during sports practice, thus helping the athlete return to full function [15].

In this context, the application of multiplanar plyometric tests can also be directed towards movement pattern analysis through the calculation of kinematic variables. Such evaluation can be performed using a markerless analysis, which allows the precise quantification of biomechanical parameters without the need for physical markers, avoiding interference during

the execution of movements. This approach has proven promising in the evaluation of various motor skills, including in the sports environment, as evidenced by recent studies that employ this type of postural estimation in soccer [16, 17]. As demonstrated by Monteiro et al (2024) [16], who analyzed kinematic differences in goalkeepers' jumps by comparing those who received video instructions with those who did not, the markerless motion capture technique identified and tracked joints and anatomical references from a 3D reconstruction of the jumps. This led to the conclusion that the instructional video generated an acute change in the analyzed movement pattern. Complementarily, Chinaglia et al. (2024) [17] also had promising results by employing instructional videos to improve the kicking kinematics of players. Both studies used the OpenPose framework. In the present study, however, this will be done using the MediaPipe framework. [18].

Markerless models, such as those implemented in the MediaPipe framework, have demonstrated good accuracy in estimating joint angles and body movements, with angular errors compatible with qualitative and quantitative analyses performed with traditional gold standard systems. This technology offers a viable, reliable, and accessible alternative compared to other traditional motion capture systems, besides increasing its applicability by allowing tests to be performed in less equipped environments. Furthermore, due to its relevance, studies have been testing its application in different environments, with a focus on athletic contexts [19, 20, 21]. For example, Dill et al. (2023) [19] demonstrated that the system presents promising results in controlled conditions, which highlights its potential as an accessible tool for physical exercise analysis. Complementarily, Fukushima et al. (2024) [21] validated the use of pose estimation in athletic and sports activities, observing angular errors compatible with qualitative performance analyses, reinforcing its viability also in field and training contexts. Van Hooren et al. (2023) [20] highlighted the effectiveness of Machine Learning in predicting ACL injury risks with high accuracy from simplified biomechanical data, suggesting that markerless systems, even with lower resolution than traditional systems, can be integrated into intelligent models for clinical and sports decision support.

The application of Machine Learning (ML) has established itself as a promising methodology for advancing research on the Anterior Cruciate Ligament, being used due to its capacity to identify risk factors [22], time to return to sport [23] and also diagnose [24]. Within the field of identifying risk factors, the study by Oliver et al. (2020) [22] employed ML algorithms to analyze data from pre-season neuromuscular assessments of 355 elite youth players and compared it with injury monitoring during the season, demonstrating that ML algorithms can identify injury risk factors with greater sensitivity than traditional regression analyses. Thus, in addition to the extraction and analysis of kinematic variables, this project proposes the use of Machine Learning algorithms as a predictive tool based on the subjective perception of rehabilitation specialists regarding the functional status of the knee. By predicting professional judgments through objective kinematic measures and the application of ML, it is expected to contribute to a more evidence-based, precise, and individualized decision-making process in athlete rehabilitation.

2 Justification

The high incidence of Anterior Cruciate Ligament (ACL) injuries [1, 2], given its importance for maintaining knee stability [1], represents a major challenge for orthopedic and sports practice.

Although widely studied, ACL rehabilitation and assessment can benefit significantly from recent advances in Machine Learning and Computer Vision [25], which have opened new possibilities for analysis, evaluation, and patient monitoring.

In this sense, this project proposes a promising approach by applying markerless Computer Vision techniques for kinematic analysis during the Multiplanar Unipodal Plyometric test associated with Machine Learning. The use of markerless systems allows for motion capture without the need for physical markers. This reduces operational costs, avoids interference with movements, and makes the methodology feasible in contexts outside the traditional clinical environment. Therefore, current literature has investigated and highlighted its effectiveness [19, 20, 21], with findings that strengthen the notion that, with proper calibration and consideration of usage contexts, markerless approaches constitute a practical, scalable, and scientifically relevant alternative in human movement analysis.

Furthermore, the use of Machine Learning models trained to predict functional assessments from movement data enables the standardization and automation of the evaluation process. This makes the analysis more precise, reduces variability among specialists, and optimizes the time of the professionals involved, offering a valuable complementary resource in recovery monitoring.

The research therefore seeks to contribute to the emerging literature related to the integration of rehabilitation, biomechanics, and technologies based on Machine Learning and Computer Vision. With this, it is intended to collaborate on the development of more accessible, objective, and scalable tools that improve functional assessment protocols in patients with Anterior Cruciate Ligament injury.

2.1 Justification for the Research Internship Abroad (BEPE)

The research internship at the University of North Florida (UNF), under the supervision of Prof. Dr. Guilherme Manna Cesar, is a crucial and strategic step for the development of this project. Prof. Cesar is an international reference in clinical biomechanics and movement analysis, with vast experience in functional assessment methodologies that are central to this study.

The infrastructure available at UNF and the integration with the local research group will allow:

1. **Cross-Cultural Validation:** Application of the evaluation protocol with US specialists (physicians and physical therapists), as the same protocol will have already been applied to the Brazilian specialists. This will allow for a comparison between clinical criteria used in Brazil and the USA, validating the model's generalizability.

2. **Methodological Refinement:** Refinement of markerless data collection and processing protocols, leveraging the host laboratory's, Laboratory of Applied Biomechanics and Engineering (LABE), expertise to validate and calibrate measurements made in Brazil.

3. **Advanced Data Analysis:** Deepening in data analysis techniques and biomechanical interpretation, with access to software and theoretical discussions that will enrich the modeling of Machine Learning algorithms.

4. **Internationalization:** Strengthening collaboration between USP and UNF, opening doors for future partnerships and high-impact joint publications.

This experience will provide a qualitative leap in the student's training and the final quality

of the project, adding specific knowledge that could not be obtained with the same depth only in the local environment.

3 Objectives

3.0.1 General Objectives

To apply predictive models based on machine learning and computer vision to estimate the functional assessment of specialists (physicians and physical therapists), via kinematic data obtained by markerless tracking during the execution of the Multiplanar Unipodal Plyometric test in patients with Anterior Cruciate Ligament (ACL) injury.

3.0.2 Specific Objectives

1. Extract and identify relevant kinematic features from the Multiplanar Unipodal Plyometric test (CUBE test) to predict the assessment of post-ACL injury recovery status.
2. Collect global functional assessments performed by specialists (physicians or physical therapists) who follow the patients, reflecting the functional level, assigning a score from 1 to 5 based on the analysis of videos of the Multiplanar Unipodal Plyometric test (CUBE) and the patient's prior clinical knowledge.
3. Train and evaluate multiple supervised machine learning models to infer the functional scores (scale of 1 to 5) assigned by Brasilian and US specialists to patients with ACL injury, using kinematic variables extracted from videos of the Multiplanar Unipodal Plyometric test (CUBE) through markerless tracking.
4. Test the transcultural applicability and generalization of the predictive model by collecting functional assessments from US-based clinical specialists (physicians or physical therapists), using these scores as an external validation set for the trained machine learning model.

3.0.3 Specific Objectives of the BEPE Internship

1. Replicate the specialist evaluation protocol with 4 US specialists (physicians or physical therapists) using the same standardized videos collected in Brazil.
2. Train and test Machine Learning models using the evaluations from US specialists to verify the generalizability and applicability of the approach in a different clinical context.
3. Compare the predictive performance of models trained with Brazilian specialists versus US specialists.
4. Discuss and refine the features selected for the Machine Learning models with Prof. Dr. Guilherme Manna Cesar and his team.
5. Write, in collaboration with the international supervisor, a scientific article with the obtained results for submission to a high-impact journal.

4 Materials and Methods

4.1 Participants

The initial estimated number of participants is between 15 and 30, based on previous studies [5, 26, 27] that used conventional statistical methods in similar research. However, considering that this project uses Machine Learning techniques, the final sample size may vary as needed to achieve an accuracy of approximately 80% in predicting the classifications made by specialists.

Individuals aged between 18 and 35 years, with a recent diagnosis of Anterior Cruciate Ligament (ACL) injury, submitted to reconstructive surgery and subsequent rehabilitation process, will participate. Inclusion criteria will be defined based on clinical history and diagnosis confirmed by the collaborating orthopedic surgeon, who will personally recruit the patients in Brazil.

This project has already been submitted to the Research Ethics Committee (Comitê de Ética em Pesquisa - CEP) via Plataforma Brasil and has been fully approved under CAAE No. 89943725.2.0000.5659 (CAAE: Certificado de Apresentação para Apreciação Ética – Brazil's National Ethics Review System Registry). All activities will be conducted in strict adherence to the approved protocol, and all participants must sign the Free and Informed Consent Form (FICF).

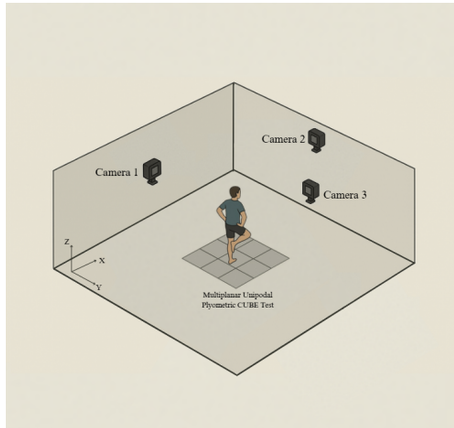
4.2 Instruments

For data collection, three GoPro Hero 10 Black cameras will be used, configured to capture videos at 2.7K resolution and 240 frames per second (fps). Filming will take place at the private office of a collaborating physician in Ribeirão Preto, Brazil. The cameras will be positioned on tripods to obtain two frontal views (a superior view and an inferior view) and one sagittal view of the patient during the test execution (Figure 1). To allow tracking in the videos and enable three-dimensional reconstruction, an object of known size will be used for environment calibration.

The choice of 2.7K resolution at 240 fps is due to the need to capture fast movements with high temporal precision, ensuring detailed kinematic analysis of patients' movement patterns. The GoPro Hero 10 Black offers this combination of resolution and frame rate, which is ideal for studies requiring minute observation of fast movements.

It is important to note that, although 2.7K resolution at 240 fps provides high-quality data for analysis, this configuration results in video files of considerable size. Therefore, adequate storage and computational capacity for processing these data will be guaranteed.

Figure 1: Camera arrangement in the office



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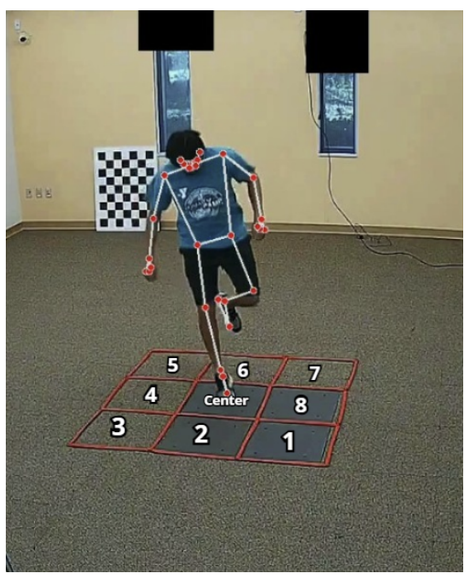
The CUBE test ("Chameleon 360 Sports Training, Inc."; Athens, AL) will be used, described as a multiplanar plyometric test of bilateral performance designed for evaluation during changes of direction. During execution, the patient performs a sequence of single-leg plyometric jumps in clockwise and counter-clockwise directions with both legs as shown in Figure 2.

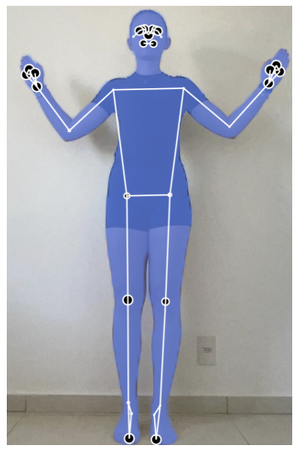
Figure 2: References for the Multiplanar Unipodal Plyometric CUBE test



Source: Created by the author (2025)

For kinematic tracking and 3D reconstruction, the *MediaPipe* multimodal framework [28] integrated into the *vailá - Multimodal Toolbox* platform [29] will be used to build Machine Learning pipelines in a user-friendly open-source manner, focusing on video and image processing. The markings will use 14 anatomical structures as references: head, trunk, arms, forearms, hands, thighs, legs, and feet. From this, 20 *landmarks* will be used, based on the 33 anatomical points model of *MediaPipe*: nose, right and left shoulders, right and left elbows, right and left wrists, tips of right and left index fingers, right and left hips, right and left knees, right and left ankles, right and left heels, and right and left big toes (hallux) (Figure 3).

Figure 3: Anatomical references for the *MediaPipe* framework.



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4.3 Evaluation Protocol for Specialists

The subjective evaluation of knee stability will be conducted using metrics that consider the risk of Anterior Cruciate Ligament (ACL) injury. Although this risk is multifactorial, several studies highlight the influence of factors such as knee valgus [3, 4, 5, 6], ankle flexion [7], and trunk, hip, and knee positions [8, 9] on joint instability and consequent increase in injury risk.

Movements with less joint flexion, close to full extension, compromise the ability of segments to adequately distribute forces during movements, resulting in knee overload. This overload is recognized as a high-risk factor for ACL injuries [10].

Clinical instruments such as the Single-Leg Landing Error Scoring System (SL-LESS) [30] and the Qualitative Analysis of Single Leg Loading (QASLS) [31] serve as references for kinetic knee evaluation, considering the previously mentioned variables. The SL-LESS, adapted from LESS for single-leg jumps, aims to identify movement patterns associated with ACL injury risk from videos in the frontal and sagittal planes [30]. The QASLS, in turn, uses distinct criteria to evaluate recordings of single-leg tasks, such as jumps and squats [31].

4.3.1 Brazilian Specialists

The evaluators will be 2 physicians and 2 physical therapists, who will have access to standardized videos of the CUBE multiplanar plyometric test, along with general project instructions. Movement evaluation will be based on their clinical references, focusing on joint stability and movement quality, especially regarding the previously highlighted risk factors. Furthermore, these variables will be considered in the development of the Machine Learning model, allowing the network to be trained with the obtained data.

To ensure result reliability, a standardized evaluation form will be developed. Calibration sessions will be conducted among specialists to align scoring criteria and minimize inter-rater variability. Agreement analysis between evaluators will be performed using Cohen's Kappa coefficient (K) [32], which measures the degree of agreement between observers. K values greater than 0.8 indicate a high level of agreement.

4.3.2 US Specialists (BEPE Phase)

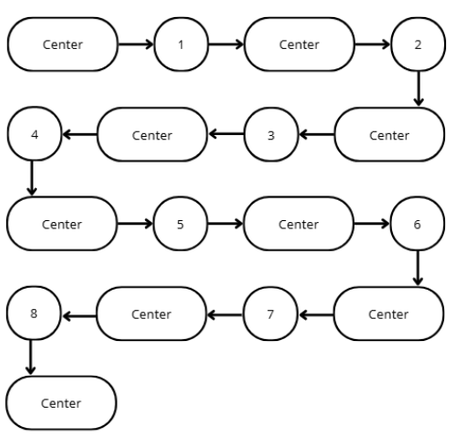
During the internship at UNF, the same protocol will be replicated with 4 US specialists (physicians and physical therapists). They will analyze the same set of standardized videos collected in Brazil. This will allow for:

1. **Cross-Cultural Comparison:** Assessing if there are significant differences in clinical judgment between Brazilian and US specialists.
2. **Model Generalization:** Training separate Machine Learning models using US specialist labels to verify if the automated system can adapt to different clinical standards or if the "ground truth" is consistent across borders.

4.4 Experimental Procedures

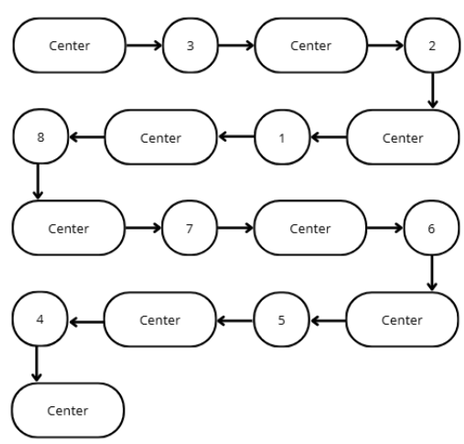
The Multiplanar Unipodal Plyometric *CUBE* test ("Chameleon 360 Sports Training, Inc."; Athens, AL) will be applied by delimiting a square on the ground segmented into 3x3 dimensions (each smaller square with a lateral measure of 0.45m) in which the patient will perform single-leg jumps in four different applications. In the first, the jumps will occur in a clockwise direction following Flowchart 1 represented by Figure 4 based on the model represented by 3. The second sequence is given by reversing the rotation direction, following Flowchart 2 of Figure 5, so patients will perform the protocol in a counter-clockwise direction. The following applications follow the same reasoning, however, there is a change of the limb with which the jumps are performed. All patients will initially perform the protocol with the right leg in the clockwise direction.

Figure 4: CUBE Flowchart - clockwise



Source: Created by the author (2025)

Figure 5: CUBE Flowchart - counter-clockwise



Source: Created by the author (2025)

Before the start of the evaluation, the test execution will be explained and exemplified; each participant will have the right to perform one attempt for better understanding and adaptation to the test. Furthermore, 4 valid samples will be collected, which can be invalidated by falls or inability to complete the test. The test and retest will be performed at three distinct moments: 1) before surgery, whenever possible; 2) months after surgery, when the patient is considered fit to perform jumps, according to the clinical criteria of the responsible specialist; and 3) approximately one year after discharge or end of the rehabilitation process, as determined by the physical

therapist. It is worth noting that these periods are general parameters, but may vary according to the rehabilitation process and the guidelines of the professionals monitoring each patient. The evaluations will be implemented as part of the analysis protocol of the partner physician of the project, integrating into the clinical follow-up of each case.

4.5 Data Processing

For kinematic analysis, *MediaPipe* will be used, an open-source framework developed by Google that allows the identification of joints and anatomical points in videos through machine learning-based skeleton detection algorithms (Figure 2). MediaPipe integrates artificial neural network models to perform real-time inference, facilitating the extraction of kinematic data from videos.

The 2D data of the obtained anatomical points will be smoothed with a fourth-order *Butterworth* digital filter, with cutoff frequency determined by residual analysis [33]. Then, the 2D coordinates will be transformed into global 3D coordinates using the 3D-DLT (*Direct Linear Transformation*) method. From the filtered and reconstructed data, kinematic variables quantifying performance in the Multiplanar Unipodal Plyometric CUBE test will be calculated.

For jump analysis, the start will be considered the moment the support foot leaves the "Center" position for the first time and, as the end, the moment it returns to this position for the last time. Jump performance analyses will be done by identifying variables at the instant the participant touches each numbered square, considering the moment of maximum flexion of the ankle, knee, and hip joints. It is important to emphasize that those responsible for data processing and machine learning model development will be blinded regarding which leg is injured, to avoid possible interference in model training and validation.

4.6 Machine Learning

This study aims to address Specific Objective 3 through the application of supervised Machine Learning techniques, with the purpose of predicting the subjective functional assessment assigned by specialists to patients with Anterior Cruciate Ligament injury. The prediction will be performed based on the analysis of kinematic variables extracted during the execution of the Multiplanar Unipodal Plyometric CUBE test. As described in topic 4.3 of this project, these variables are associated with increased ACL injury risk and include: knee valgus, trunk flexion, hip flexion, knee flexion, and ankle flexion.

In the BEPE phase, this process will be expanded to include training and validation with data labeled by US specialists, allowing a direct comparison between models trained with Brazilian vs. US clinical opinions.

For the development of predictive models, the *Scikit-learn* library will be used, an open-source tool in Python widely adopted for supervised machine learning tasks, including classification, regression, and clustering [34]. *Scikit-learn* provides a consistent and efficient interface for the implementation of various classification algorithms, facilitating experimentation and model comparison.

The performances of the following supervised classification algorithms will be evaluated and compared:

1. **Extreme Gradient Boosting (XGBoost)**: a machine learning algorithm based on decision trees, known for its efficiency and performance in data science competitions [35].

2. **Random Forest**: an ensemble method that builds multiple decision trees and combines them to obtain a more precise and stable prediction [36].

3. **Logistic Regression**: a statistical model that uses a logistic function to model the probability of a certain class or event [37].

4. **K-Nearest Neighbors (KNN)**: an algorithm that classifies a sample based on the classes of its nearest neighbors in the feature space [38].

5. **Support Vector Machine (SVM)**: a model that finds the optimal hyperplane separating classes in the feature space, being effective in high-dimensional spaces [39].

6. **Gaussian Naive Bayes**: a probabilistic classifier based on Bayes' Theorem, assuming independence between features [40].

7. **Multilayer Perceptron (MLP)**: a feedforward neural network composed of multiple layers of neurons, capable of modeling complex non-linear relationships [41].

It is important to highlight that, although Machine Learning techniques generally require large volumes of data for effective model generalization, this study faces limitations such as the time available for data collection and analysis, in addition to being a scientific initiation project. Consequently, the obtained sample may not fully represent the complexity needed for building highly generalizable models. However, this study constitutes an initial step that can be expanded and deepened in future investigations, with the inclusion of a larger volume of data and, consequently, the improvement of the developed predictive models.

4.6.1 Pre-Processing

The kinematic variables extracted from the CUBE test will be organized and combined to build the datasets used as predictors in the machine learning models. The variables will be normalized using the *StandardScaler* from the Scikit-learn library, ensuring zero mean and unit variance in each feature.

Hyperparameter selection and optimization will be performed using the *Grid Search* technique, adopting balanced accuracy obtained in validation sets as the main criterion. To ensure that class proportion is maintained in all partitions, the *StratifiedKFold* technique will be used during cross-validation, ensuring the representativeness of minority classes and greater robustness in model evaluation.

To ensure experiment reproducibility, a fixed *random_state* value will be used in all data splits. Cross-validation will be repeated 30 times, employing different random splits, to reduce result variability and provide a robust evaluation of model performance.

4.6.2 Model Evaluation

Machine learning model evaluation will be conducted using the following metrics: balanced accuracy, precision, recall, and F1-Score. These metrics will be calculated both individually for each class and for the overall average (*macro average*), allowing a comprehensive analysis of model performance in different classification scenarios.

Finally, the combination of kinematic variables and hyperparameters presenting the best performance in balanced accuracy for predicting the functional assessment assigned by

specialists will be selected. From this approach, we seek to identify movement patterns that significantly contribute to predicting the knee stability status of patients evaluated during the CUBE test.

4.7 Statistical Analysis

Performance metrics of Machine Learning models, as well as values of kinematic variables extracted from the Multiplanar Unipodal Plyometric CUBE test, will be described using mean and standard deviation. Comparisons will be made between the best-performing models and between different evaluation moments of participants.

Data normality verification will be conducted using the Shapiro-Wilk test, while variance homogeneity will be evaluated by Levene's test. In cases where data present normal distribution and homogeneous variances, one-way analysis of variance (ANOVA) will be applied, followed by Tukey's HSD (Honestly Significant Difference) post-hoc test for multiple comparisons.

When normality or homogeneity assumptions are not met, the Kruskal-Wallis non-parametric test will be used, followed by Dunn's post-hoc test with Bonferroni adjustment for group comparisons. Effect size for each comparison will be estimated by Cohen's d.

The significance level adopted for all analyses will be $p < 0.05$. All statistical analyses will be performed using Python 3.12.9 codes, employing the libraries `SciPy` [42], `statsmodels` [43], and `scikit-posthocs` [44].

5 Activities to be developed by the scholarship holder

The project execution will be structured into two complementary phases to fulfill the objectives of the Scientific Initiation (IC) and the Research Internship Abroad (BEPE), ensuring the acquisition of proficiency in advanced collection, processing, and cross-cultural validation techniques for predictive models.

5.1 National Phase (Brazil - USP)

This phase is dedicated to the acquisition of primary data, protocol organization, and preparation of the dataset for advanced analysis at the host institution, concluding before the start of the BEPE fellowship.

1. **Literature Review:** Conduct an in-depth review of the literature on ACL injury biomechanics, markerless systems (MediaPipe), and the application of Machine Learning in rehabilitation.

2. **Experimental Preparation:** Configure and adjust instruments (GoPro cameras and tripods), and perform a pilot study to refine the video collection protocol.

3. **Primary Data Collection:** Recruit and follow up with participants, performing the Multiplanar Unipodal Plyometric CUBE test at the three specified moments of evaluation.

4. **Data Processing and Feature Extraction:** Apply video processing techniques using the MediaPipe framework for 2D coordinate extraction, 3D transformation (DLT), and data filtering, culminating in the calculation of relevant kinematic variables (features).

5. **Collection of National Labels:** Organize and collect the global functional assessments (scale of 1 to 5) from Brazilian clinical specialists (physicians and physical therapists), which will serve as the primary label for training the Machine Learning model.

6. **Machine Learning Model Training and Optimization:** Train, optimize, and evaluate multiple supervised Machine Learning models using the Brazilian specialist assessments.

5.2 International Phase (United States of America - BEPE Internship at UNF)

This phase will be conducted under the supervision of Dr. Guilherme Manna Cesar at the University of North Florida, focusing on advanced data processing, supervised Machine Learning model training, and international validation.

1. **Collection of International Labels:** Replicate the specialist evaluation protocol by collecting functional scores from US-based clinical specialists (physicians or physical therapists) using the same standardized videos collected in Brazil.

2. **Transcultural Generalization Analysis:** Test the transcultural applicability and generalization of the predictive model by utilizing the scores from the US specialists as an independent external validation set, and discuss and refine the selected features with the host supervisor and his team.

3. **Scientific Production and Reporting:** Write, in collaboration with the international supervisor, a scientific article detailing the methodology and results for submission to a high-impact journal, and prepare the Final BEPE Report .

6 Collaborators

The project counts on the collaboration of Dr. Rodrigo Salim, co-advisor and direct stakeholder in the research, responsible for inviting and referring patients for evaluations. Given his recognized professional and academic experience, his role is fundamental to the project's success, ensuring clinical rigor in the selection and follow-up process of participants.

As a collaborating doctoral student, Rafael Luiz Martins Monteiro participates, contributing actively to experimental development, data analysis, and discussion of results, expanding integration between different levels of academic training and strengthening the multidisciplinary character of the research.

Furthermore, the project features the international collaboration of researcher Dr. Guilherme Manna Cesar, from the University of North Florida (UNF), who will serve as the host supervisor during the BEPE fellowship phase. His role is essential for guiding the advanced data processing, the machine learning analysis, and the cross-cultural validation protocol with US specialists at UNF, thereby expanding opportunities for academic exchange, scientific collaboration, and the enrichment of the training for those involved in the project.

7 Schedule

The presented schedule distributes the main stages of the project over four months, organizing from adaptation and integration into research group to discussion of results, and finalizing with article writing and final BEPE report. This sequence ensures that each stage is conducted in a structured way, favoring the achievement of the study objectives within the foreseen deadline.

Table 1: Schedule of Activities for BEPE Internship (Start: 12/01/2026)

Activities	**Months (2026-2027)**			
	December	**January**	**February**	**March**
Adaptation and Integration into Research Group				
Training in Advanced Methodologies				
Recruitment of US Specialists				
Data Collection (US Specialist Ratings)				
Refinement of ML Models (US Data)				
Discussion of Results and Article Writing				
Preparation of Final BEPE Report				

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